

mortgage comes with advantages, the biggest being the home is paid off much sooner, and equity is built much more quickly. Mortgage interest rates are often higher for 30-year mortgages than 15-year mortgages. However, the payments for 15-year mortgages are considerably higher. The following exercises explore the difference between a 15- and 30-year mortgage for a mortgage of \$200,000.

39. The 15-year mortgage interest rate is 5.6%.
 - a. Find the payment.
 - b. Determine the total that would be paid if the mortgage was completed.
 - c. Find the cost of financing for this mortgage.
40. The 30-year mortgage rate is 6.25%.
 - a. Find the payment.
 - b. Determine the total that would be paid if the mortgage was completed.
 - c. Find the cost of financing for this mortgage.
41. How different are the payments, the total paid, and the cost to finance?
42. Summarize the answer in the previous question.

6.13 Income Tax



Figure 6.28 Federal income tax is a concern for most US citizens. (credit: "1040 US tax form" by Marco Verch Professional Photographer/Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Determine gross, adjusted gross, and taxable income.
2. Apply exemptions, deductions, and credits to basic income tax calculations.
3. Compute FICA tax.
4. Solve tax application problems for working students.

Before the start of the American Civil War in 1861, most of the country's revenue came from tariffs on trade and excise taxes. However, this fell far short of the high cost of the war. Because of this, the federal government enacted the nation's first income tax with the Revenue Act of 1861, which created the Internal Revenue Service as we know it today.

No one likes paying income tax, but it is a reality of life. In this section, we will learn about Form 1040, the U.S. Individual Income Tax Return, and ways to prepare for tax time.

The U.S. tax code may change from year to year. Because of this, this section includes examples of how taxes, deductions, and exemptions might be computed. The types of income, deductions, and exemptions that are used in the examples are used in the current tax code.

Gross, Adjusted Gross, and Taxable Income

Your income drives how much you pay in taxes. The more you earn, the more you are likely to pay. But your income alone is not the full story. When you add all the money you earned from your job, freelance work, interest from savings, and other sources, you have your **gross income**. If you are an employee, your income from your job will be reported on a W-2, which is sent to you by your employer. Income from freelance work will be reported on a 1099-MISC form, and is sent by the company that paid you. Income from interest is reported on a 1099-INT form and comes from the entity that paid the interest.

Before you determine how much you owe in taxes, you will make certain adjustments to that gross income. You will deduct, or subtract, some of income from the gross income. That's your **adjusted gross income**, or AGI. That is still not what you are taxed on. Next, you need to apply **exemptions** to your income. These are pieces of income that the government does not tax. After that is done, you reach your **taxable income**. We will look at each of these parts of the taxable income.

? WHO KNEW?

Gifts and Winning

Money given as a gift may be taxed if the gift amount is high enough. If you win \$50,000 in the lottery, that money is taxed as income. If you give a family member a large cash gift, that gift will be subject to a tax provided that the gift exceeds the federally set limits.

You will notice that your paycheck already has taxes taken out of it. Your employer will withhold some of your income, sending it directly to the federal, state, and local governments. It is an estimate of how much you will owe in income tax. In the end, it reduces how much you will pay when your taxes are due. If they withhold too much income, you will receive the extra they withheld in the form of a refund.

EXAMPLE 6.115

Computing Gross Income

Roger is preparing to do his taxes. He worked two jobs, with reported income of \$27,500 and \$13,200. His two CDs yielded \$327 together. He also won a split club raffle for \$2,000. What was Roger's gross earnings?

✓ Solution

This is the sum of his wages, winnings and interest earned. Add these Roger's gross income was \$43,027.

> YOUR TURN 6.115

1. Chloe prepares for her taxes by collecting her W-2 and her 1099-MISC documents. The wages from her regular job were \$41,780. She did some freelance work and earned \$5,500 from that job. What is Chloe's gross income?

Your adjusted gross income (AGI) is computed before your taxes are determined. It begins with the gross income, and then subtracts from that income any deduction. **Deductions** are expenditures on your part that the government won't tax. These deductions include money deposited into tax-deferred investments, and mortgage interest that you paid, charitable contributions if you made any, medical bills over a threshold, medical insurance under certain circumstances, and property taxes. If you add all these up, and they are all legal deductions, the sum is subtracted from your gross income, leaving the AGI.

EXAMPLE 6.116

Compute Adjusted Gross Income

Chandra's gross income is \$58,400. She deposited \$3,000 into her tax-deferred retirement account for the year. Her mortgage interest paid was \$1,250 for the year, and her property taxes came to \$4,200. What is her AGI?

✓ Solution

To get Chandra's AGI, total her deductions and subtract that total from her gross income. The total of her deductions is $\$3,000 + \$1,250 + \$4,200 = \$8,450$. She subtracts that total from her gross income, making her AGI $\$58,400 - \$8,450 = \$57,950$.

> YOUR TURN 6.116

1. Isabelle's gross income is \$93,450. She paid \$7,840 in mortgage interest and \$3,810 in property taxes. She also donated \$1,500 to her favorite charity. What is Isabelle's AGI?

Remember that your AGI is not your **taxable income**. Exemptions need to be subtracted from the AGI to reach your taxable income. Exemptions are income that the government does not tax. Some examples of exempt income are disbursements from health savings accounts for qualified medical expenses, bond interest, some IRA distributions, and gifts given that are under \$16,000. Note that exemptions are different from deductions: exemptions are excused incomes, whereas deductions are excused expenditures.

EXAMPLE 6.117

Taxable Income

Yelizaveta's AGI is \$75,490. However, she gave a \$2,000 gift to her mother. What is her taxable income?

✓ Solution

Taxable income is AGI minus any exemptions. Gifts below \$16,000 are exempted, so her taxable income is $\$75,490 - \$2,000 = \$75,490$.

> YOUR TURN 6.117

1. Ammie has an AGI of \$43,100. However, \$3,400 was a disbursement from her health savings account. What is her taxable income?

Tax Credits

Another piece of the tax puzzle is **tax credits**. This is money subtracted from the tax you owe.

Tax credits are very different from deductions or exemptions. Deductions and exemptions are taken away from your gross income before the tax you owe is calculated. A tax credit, is subtracted, dollar for dollar, from your tax bill. Once the tax you owe is calculated, subtract the any tax credits from that calculated tax.

Some of the tax credits are refundable. This means that if subtracting them from your tax results in a negative number, you receive a tax refund. For more details, see [this article about tax credits \(https://openstax.org/r/taxcredit.asp\)](https://openstax.org/r/taxcredit.asp).

The federal government has placed income limits and restrictions and on those eligible to receive tax credits because their value is so high. Here is a partial list of tax credits that you might qualify for:

- **Earned income credit** is a refundable tax credit for low- to moderate-income workers and ranges from \$560 to \$6,935 depending on dependents and income. This is refundable
- **American opportunity credit** is a credit taken by parents who have children enrolled in college at least half time and pursuing a degree. This credit is worth \$2,500 per student for the first 4 years of undergraduate school, subject to income limits. This is a refundable tax credit.
- **Lifetime learning credit** is a credit is equivalent to 20% of educational expenses, up to \$2,000 per year, subject to income limits. There is no cap to how many years you can apply for this credit.
- **Child tax credit** is worth \$2,000 per child under the age of 17 if that child lives at home at least half the year, subject to income limits. This is a refundable tax credit.
- **Child and dependent care tax credit** was designed to help pay for child care while the parent works. The amount of the credit is dependent on your income. However, the maximum amount that can be received is, in 2022, \$4,000 for one eligible person, or \$8,000 for two or more qualifying people. A dependent qualifies if they are a child under

13 years old, a spouse who is unable to care for themselves, or some other qualifying person. This is a refundable tax credit.

- **Premium tax credit** was created by the Affordable Care Act, and it is one that is received by many people throughout the year. In essence it is a health insurance premium subsidy. The amount of the credit is based on your income and the price of health insurance in your area. This is a refundable tax credit.

EXAMPLE 6.118

Apply a Tax Credit

Kaitlyn has calculated the tax she owes to be \$5,200. However, she receives an earned income tax credit of \$1,715. How much does Kaitlyn owe after applying the earned income tax credit?

Solution

The amount of taxes Kaitlyn owes is her calculated tax of \$5,200 minus the credit she receives. The amount she owes in taxes is $\$5,200 - \$1,715 = \$3,485$.

YOUR TURN 6.118

1. Antonio's taxes owed based on his income are \$3,950. However, he qualifies for the earned income tax credit of \$1,925 and a child tax credit of \$2,000. How much does Antonio owe after applying the tax credits?

EXAMPLE 6.119

Tax Credits and a Refund

Chanajah calculated their tax owed, which came to \$4,300. They have an earned income tax credit of \$2,190, a child tax credit of \$2,000, and a child and dependent care tax credit of \$4,000. How much tax does Chanajah owe, or how much will they get in a refund?

Solution

Adding Chanajah's tax credits together, we find their total to be \$8,190. That is more than the tax they owe, which was \$4,300. Each of those credits is refundable, which means they will receive a refund. Subtracting the credits from the tax owed yields $\$4,300 - \$8,190 = -\$3,890$. This is negative, so represents a refund of \$3,890.

YOUR TURN 6.119

1. Ismail owed \$2,350 in taxes. He has a child tax credit of \$2,000 and an earned income tax credit of \$1,640. After applying the tax credit, how much does Ismail owe or how much does he get in a refund?

VIDEO

[Deductions Versus Credits \(https://openstax.org/r/Versus_Credits\)](https://openstax.org/r/Versus_Credits)

Computing FICA Taxes

FICA stands for the Federal Insurance Contributions Act of 1935. FICA taxes are used solely to fund Social Security and Medicare and are separate from federal income tax. It amounts to 7.65% of your gross pay, which is withheld from your paycheck automatically. Your employer is required to match the 7.65% amount. Of the 7.65%, 6.2% goes to Social Security (SSI), and 1.45% goes to Medicare.

As of 2022, SSI tax only applies to the first \$147,000 of earnings. Any gross income above that is not taxed for social security. This limit changes every year.

Medicare tax, on the other hand, applies to the entirety of your gross income.

EXAMPLE 6.120**Computed FICA Taxes**

McKenzi earned \$2,700 in gross income, before taxes, in a given 2-week period. How much does she owe in FICA taxes, and how much of that is for SSI?

✓ **Solution**

The FICA tax is 7.65% of her gross earnings. 7.65% of her \$2,700 is \$206.55. Also, the SSI is 6.2% of her income, or \$167.40

> YOUR TURN 6.120

1. Arianne earned \$3,200 over a 2-week pay period. How much FICA tax does she pay, and how much of that is for SSI?

EXAMPLE 6.121**Social Security Tax with Higher Income**

Renard earned \$195,000 in wages for the year. How much in SSI taxes does Renard owe for the year?

✓ **Solution**

Since Renard earned more than the taxable limit of \$147,000 dollars, he only pays the 6.2% SSI tax on \$147,000. This comes to \$9,114.

> YOUR TURN 6.121

1. Andy earned \$169,450 in wages this year. How much in SSI taxes does Andy owe?

Calculating Your Income Tax

Your income tax bill and your income tax rate are based on your taxable income. The tax system in the United States is progressive, meaning that the tax rates are marginal so the higher your taxable income the higher the tax rate you will pay. Taxable income is broken into brackets, or ranges of income. Each bracket has a different tax rate. The tax brackets and rates for single filers as of 2022 are given below:

Bracket	Lower Income Limit	Upper Income Limit	Tax Rate
1	0	\$10,275	10%
2	\$10,276	\$41,775	12%
3	\$41,776	\$89,075	22%
4	\$89,076	\$170,050	24%
5	\$170,051	\$215,950	32%

Table 6.3 Federal Income Tax Brackets for Single Filers, 2021–2022
 (data source: <https://www.irs.gov/newsroom/irs-provides-tax-inflation-adjustments-for-tax-year-2022>)

Bracket	Lower Income Limit	Upper Income Limit	Tax Rate
6	\$215,951	\$539,900	35%
7	\$539,901		37%

Table 6.3 Federal Income Tax Brackets for Single Filers, 2021–2022
(data source: <https://www.irs.gov/newsroom/irs-provides-tax-inflation-adjustments-for-tax-year-2022>)

So if your taxable income is \$76,500 and you are filing as a single filer, your tax bill will be 22% of that \$76,500, right?

Wrong. Your income is split among those brackets and the money in each bracket is taxed at that bracket's tax rate. Seems confusing. Here is a list of steps to follow to find the tax owed.

Step 1: Find the bracket for the income.

Step 2: For each bracket below the income bracket, the tax from that bracket is:

Step 2a: Find the difference between the upper limit of that bracket and upper limit of the next lower bracket. If this is bracket 1, use 0 as the upper limit of the previous bracket.

Step 2b: The tax from that bracket is the bracket tax rate applied to the difference from Step 2a.

Step 3: For the bracket that the income belongs to, find the income minus the lower limit for the bracket.

Step 4: The tax for the bracket of the income is tax rate for that bracket applied to the difference found in Step 3.

Step 5: Add these various tax values to get the total income tax.

There are various tax brackets, and the rates may change in any given year. The income limits may also change. For all examples going forward, we will use the single filer tax brackets, even if those brackets are not appropriate (e.g., married or head of household filers).

EXAMPLE 6.122

Income Tax on Taxable Income

Faith has a taxable income of \$103,650. How much income tax does Faith owe?

Solution

Step 1: Faith's income belongs to the 4th tax bracket, \$89,076 to \$170,050. So the taxes from the first three brackets follow steps 2a and 2b.

For bracket 1:

Step 2a: The upper limit is \$10,275, the upper limit of the previous bracket was 0, so the difference is \$10,275.

Step 2b: For the first bracket, she owes 10% of \$10,275, or \$1,027.50.

For bracket 2:

Step 2a: The upper limit is \$41,775, the upper limit from the previous bracket was \$10,275, so the difference is \$31,500

Step 2b: For the second bracket, she owes 12% of \$31,500, or \$3,780.

For bracket 3:

Step 2a: The upper limit is \$89,075, the upper limit from the previous bracket is \$41,175, so the difference was \$47,300

Step 2b: For the third bracket, she owes 22% of \$47,300, or \$10,406.

Step 3: For bracket 4, her income is \$103,650, the upper limit of the previous bracket is \$89,075. The difference of those is $\$103,650 - \$89,075 = \$14,575$.

Step 4: The tax he owes for that bracket is 24% of the difference, which is \$3,498.

Step 5: The total in taxes that Emmanuel owes is the sum of the taxes found above, or

$$\$1,027.5 + \$3,780 + \$10,406 + \$3,498 = \$18,711.50$$

> YOUR TURN 6.122

1. Jenna's taxable income is \$73,500. Find how much income tax she owes.

 Remember that your employer will estimate how much tax you will owe and withholds it from paychecks. This means you may have already paid some, if not all and more, of the income tax you owe.

▶ VIDEO

[Tax Brackets \(https://openstax.org/r/Brackets\)](https://openstax.org/r/Brackets)

EXAMPLE 6.123

Finding Income Tax Owed

Emmanuel is preparing his taxes. His W-2 from work shows gross income for the year of \$95,250. He also has a 1099-MISC for some freelance art work he did, amounting to \$7,500. Emmanuel also deposited \$4,500 into a tax-deferred retirement plan. He paid \$7,920 in mortgage interest for the year and \$3,740 in property taxes. He also qualified for \$4,000 in child tax credits. Based on this information, how much tax does Emmanuel owe or how much does he get in a refund?

Solution

We need to know Emmanuel's taxable income, based on gross income, deductions, and exemptions.

His gross income is the amount from his W-2 and his 1099-MISC. Adding these gives a gross income of $\$95,250 + \$7,500 = \$102,750$.

Subtracting his deductions will yield his AGI. His deductions are \$7,920 for mortgage interest, \$3,740 for property taxes, and \$4,500 deposited into his retirement account. Adding these, his total deductions are \$16,160. Subtracting the deductions from the gross income, we find his AGI to be \$86,590.

Emmanuel seems to have no exemptions, so his AGI and his taxable income are the same.

We find the taxes Emmanuel owes using the process outlined above.

Step 1: His income belongs to the third tax bracket, \$41,776 to \$89,0875. So the taxes from the first two brackets follow steps 2a and 2b.

For bracket 1:

Step 2a: The upper limit is \$10,275, the upper limit of the previous bracket was 0, so the difference is \$10,275.

Step 2b: For the first bracket, he owes 10% of \$10,275, or \$1,027.50.

For bracket 2:

Step 2a: The upper limit is \$41,775, the upper limit from the previous bracket was \$10,275, so the difference is \$31,500

Step 2b: For the second bracket, he owes 12% of \$31,500, or \$3,780.

Step 3: For bracket 3, his income is \$86,590, the upper limit of the previous bracket is \$41,775. The difference of those is $\$86,590 - \$41,775 = \$44,815$.

Step 4: The tax he owes for that bracket is 22% of the difference, which is \$9,859.30.

Step 5: The total in taxes that Emmanuel owes is the sum of the taxes found above, or $\$1,027.5 + \$3,780 + \$9,859.30 = \$14,666.80$.

Once his taxes are computed, he subtracts his tax credits. His only tax credit is \$4,000. The total he owes in taxes is $\$14,666.80 - \$4,000 = \$10,666.80$.

 YOUR TURN 6.123

1. Jacob is preparing his taxes. His W-2 from work shows gross income for the year of \$39,885. He also has a 1099-INT for a savings account, amounting to \$378. Jacob also deposited \$2,500 into a tax-deferred retirement plan. He also qualified for \$2,000 in child tax credits and \$4,000 in child and dependent care tax credit. Based on this information, how much tax does Jacob owe or how much does he receive in a refund?

Check Your Understanding

86. Gross income consists of what?
87. Taxable income is found after what is done to gross income?
88. If the gross income is \$50,000 and there are three deductions of \$2,000, \$3,000, and \$8,000, what is the adjusted gross income?
89. When is a tax credit applied?
90. Do you pay 6.2% for SSI tax for all your earnings?
91. For a taxable income of \$41,800, how much income tax is owed?
92. If \$5,600 is owed in taxes but there are \$4,000 in tax credits, how much income tax is owed?



SECTION 6.13 EXERCISES

1. What is gross income?
2. How is adjusted gross income derived from gross income?
3. Is there a difference between AGI and taxable income? If so, what is the difference?
4. How are deductions different from tax credits?
5. What are the two components of the FICA tax?
6. What taxes do employers match?
7. If you are in the 32% tax bracket, is your tax computed on your full income?
8. What does it mean for a tax credit to be refundable?

In the following exercises, compute the gross income.

9. Wages from W-2 are \$32,800, freelance income from a 1099-MISC is \$1,050, and a gift given to a family member of \$2,760.
10. Wages from W-2 are \$59,380, interest income from 1099-INT is \$1,500, freelance income from a 1099-MISC is \$3,500, and a split club raffle winning of \$6,755.
11. Wages from job 1 on W-2 are \$36,200, wages from job 2 on W-2 are \$21,400, interest income from a 1099-INT is \$374.
12. Wages from W-2 are \$121,450, interest from a 1099-INT is \$3,400.

In the following exercises find the adjusted gross income, AGI, based on gross income and deductions.

13. Gross income was \$65,700, \$2,280 deposited in a tax-deferred IRA, mortgage interest was \$4,715, property taxes were \$3,065.
14. Gross income was \$183,200, \$5,000 deposited in a tax-deferred IRA, \$7,300 in charitable contributions, \$8,350 in mortgage interest, and \$6,900 in property taxes.
15. Gross income was \$31,200, \$1,500 deposited in a tax-deferred IRA account.
16. Gross income was \$41,500, \$1,250 deposited in a tax-deferred IRA account, \$4,210 in mortgage interest, \$2,980 in property tax.

In the following exercises, find the taxable income based on AGI and exemptions.

17. AGI of \$34,560, \$2,500 disbursement from a health savings account for a qualifying medical expense.
18. AGI of \$56,750, gift given to a family member of \$8,000, \$550 in bond interest.
19. AGI of \$120,940, gift given to a family member of \$15,000, bond interest of \$4,500.
20. AGI of \$28,450, gift given to a family member of \$2,000.

In the following exercises, find the taxes owed or refund received based on income tax bill and tax credits.

21. Tax bill of \$5,300, child tax credit of \$4,000, earned income tax credit of \$1,630.
22. Tax bill is \$17,300, child tax credit of \$4,000, child and dependent care tax credit \$8,000, lifetime learning credit of \$1,000.
23. Tax bill of \$5,205, child tax credit of \$6,000, earned income tax credit of \$2,450.
24. Tax bill of \$11,300, child tax credit of \$2,000, earned income tax credit of \$650, child and dependent care credit of \$4,000.
25. Tax bill of \$13,750, child tax credit of \$2,000, earned income tax credit of \$780, child and dependent care credit of \$2,000.

In the following exercises, determine the SSI tax and the total FICA tax for the given incomes.

26. Earning for pay period were \$3,500.
27. Earnings for pay period were \$1,400.
28. Earnings for pay period were \$3,150.
29. Earnings for the year were \$135,000.
30. Earnings for the year were \$203,400.

In the following exercises, find the income tax owed using the taxable income and the tax table below, for a person filing single.

Bracket	Lower Income Limit	Upper Income Limit	Tax Rate
1	0	\$10,275	10%
2	\$10,276	\$41,775	12%
3	\$41,776	\$89,075	22%
4	\$89,076	\$170,050	24%
5	\$170,051	\$215,950	32%
6	\$215,951	\$539,900	35%
7	\$539,901		37%

31. Taxable income of \$36,250.
32. Taxable income of \$63,500.
33. Taxable income of \$209,450.
34. Taxable income of \$92,250.
35. Alexandra is preparing her taxes. Her W-2 from work shows gross income for the year of \$51,300. She also has a 1099-INT for interest on a savings account for \$910. Alexandra also deposited \$2,750 into a tax-deferred retirement plan. She paid \$2,150 in mortgage interest for the year and \$2,060 in property taxes. Based on this information, how much tax does Alexandra owe or how much does she get in a refund?
36. Tymoteusz is preparing his taxes. His W-2 from work shows gross income for the year of \$47,680. He also has a 1099-MISC for some freelance consulting he did, amounting to \$1,800. Tymoteusz deposited \$3,000 into a tax-deferred retirement plan. He also qualified for \$2,000 in child tax credits and \$4,000 in child and dependent care tax credits. Based on this information, how much tax does Tymoteusz owe or how much does he get in a refund?